

Vehicle Architecture-Driven Design and Comprehensive Loss Analysis of an 800V Three-Level Flying Capacitor Traction Inverter

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Project Overview

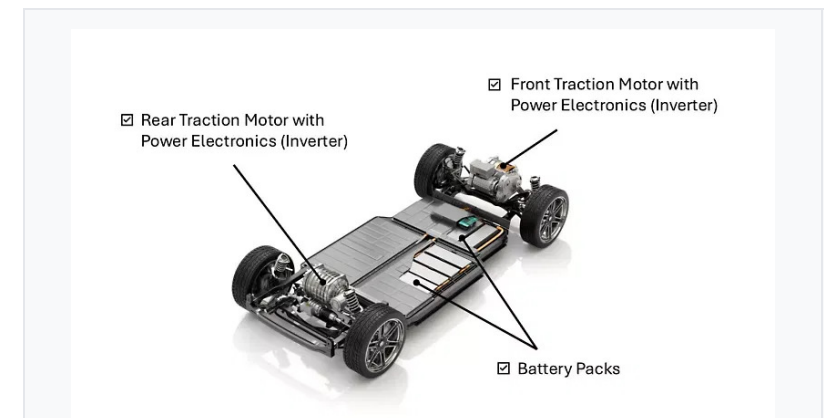
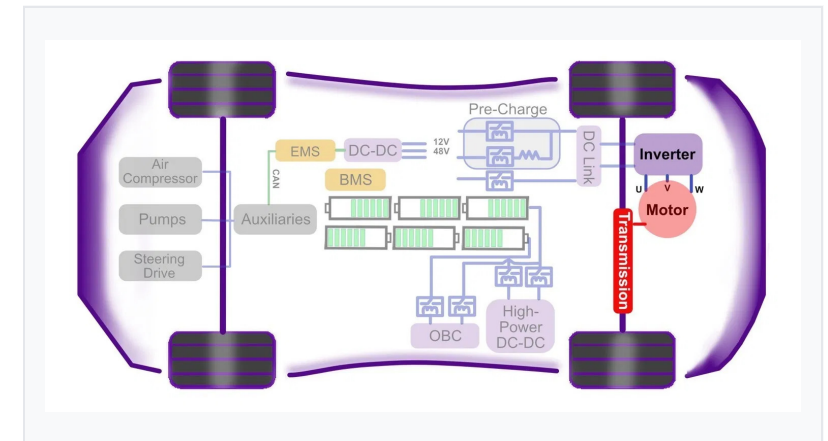
- **The Core Problem:** Traction inverter design requires balancing efficiency, torque quality, and thermal stress across highly variable driving conditions.
- **Variable Frequency:** The required electrical fundamental frequency (f_e) is not fixed; it is directly proportional to the vehicle's mechanical speed.
- **Load Profile:** Urban driving demands low frequency but high phase currents for acceleration. Highway driving demands high frequency but lower sustained currents.

Project Objective

To model a traction inverter across a full spectrum of operating conditions (urban to highway), evaluate comprehensive conduction and switching losses, and optimize the design for maximum drivetrain efficiency and thermal reliability.

Design Choice: 800V Architecture

- **The Baseline Decision:** To address the performance demands of modern EVs, an 800V DC bus architecture was explicitly chosen as the baseline for this design project.
- **Current Reduction:** Adopting an 800V architecture halves the required phase current for a given power level compared to traditional 400V systems.
- **Efficiency Gains:** This drastic reduction in current significantly lowers I^2R ohmic losses in the inverter semiconductors, motor windings, and high-voltage cabling.



The dv/dt Problem and Multilevel Solution

- **The 1200V Requirement:** Traditional two-level voltage source inverters (2L-VSI) require 1200V-class semiconductors to operate on an 800V DC bus.
- **The SiC Trade-off:** While Silicon Carbide (SiC) devices easily meet this voltage requirement and offer low losses, their extremely fast switching transitions generate severe **dv/dt stress**.
- **Motor Degradation:** This high rate of voltage change over time leads to premature motor winding insulation failure and bearing degradation.

The Multilevel Solution

- To mitigate dv/dt stress, a **multilevel inverter topology** is required.
- It synthesizes the output voltage from multiple discrete levels, inherently reducing the voltage step size (e.g., 400V steps instead of 800V steps).
- This cuts the dv/dt stress in half and significantly improves the harmonic profile of the output waveform.

Topology Selection: FC vs NPC

Thermal Symmetry & Component Count

- **Asymmetric Stress in NPC:** NPC suffers from unequal loss distribution between inner and outer switches, complicating cooling system design.
- **Symmetric Stress in FC:** FC distributes voltage and current stresses symmetrically across all four switches. Every switch blocks exactly half the DC bus voltage ($V_{dc}/2$).
- **Reduced Conduction Loss:** FC eliminates the need for clamping diodes required in the NPC, removing components from the conduction path.

Dynamic Load Balancing

- **Transient Instability in NPC:** The NPC neutral point is notoriously difficult to keep balanced under highly dynamic, rapidly fluctuating EV loads.
- **Natural Balancing in FC:** FC utilizes redundant switching states to naturally balance the flying capacitor voltage.
- **Power Factor Independence:** By actively selecting between redundant states based on the phase current direction, the FC remains balanced at $V_{dc}/2$ regardless of the load power factor.

FC Switching States and PD-PWM

- 3-Level Output:** Synthesizes $+V_{dc}/2$, 0, and $-V_{dc}/2$ using four switches per phase, reducing voltage steps and dv/dt stress.
- Redundant Zero States:** The '0' voltage level can be achieved by two different switch combinations, providing a degree of freedom in control.
- Natural Balancing:** Phase-Disposition PWM (PD-PWM) naturally balances the flying capacitor voltage without complex closed-loop control.
- Mechanism:** By alternating between redundant zero states based on the phase current direction, the capacitor charges or discharges to maintain exactly $V_{dc}/2$.

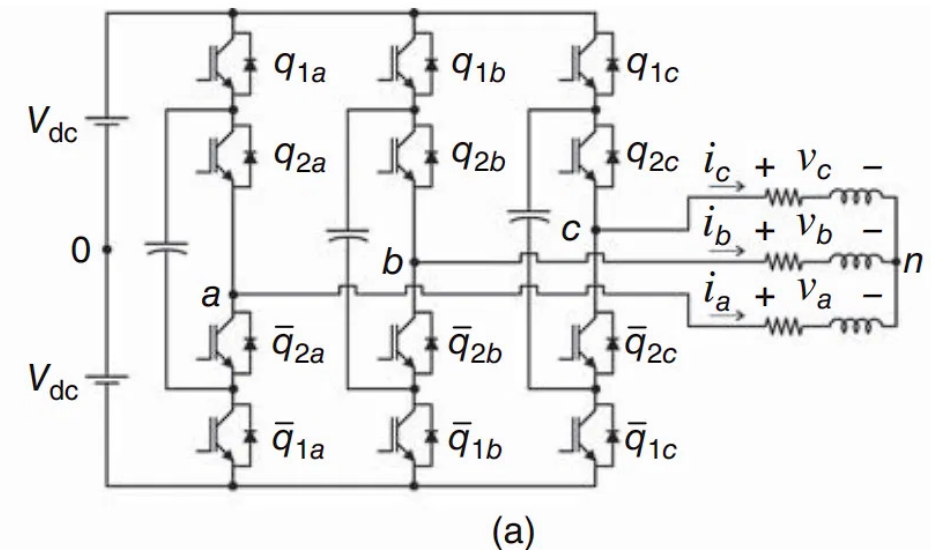


Fig. 1. Three-phase schematic diagram of the 3-level flying capacitor traction inverter.

Source: Advanced Power Electronics Converters (E. Cipriano).

Component Selection

Active Switch: SiC Module

- **Wolfspeed CAB450M12XM3:** 1200V, 450A SiC half-bridge module.
- **Voltage Margin (1200V):** Large safety margin over 400V stress; meets IPC-9592B derating at 33% use.
- **Current Margin (450A):** ~3.3x margin vs. 136A peak phase current.
- **Thermal Stability:** Positive temp. coeff. of R_{ds_on} (3.3 $\rightarrow$ 6.6m Ω) helps prevent thermal runaway.

Flying Capacitor: Film Tech

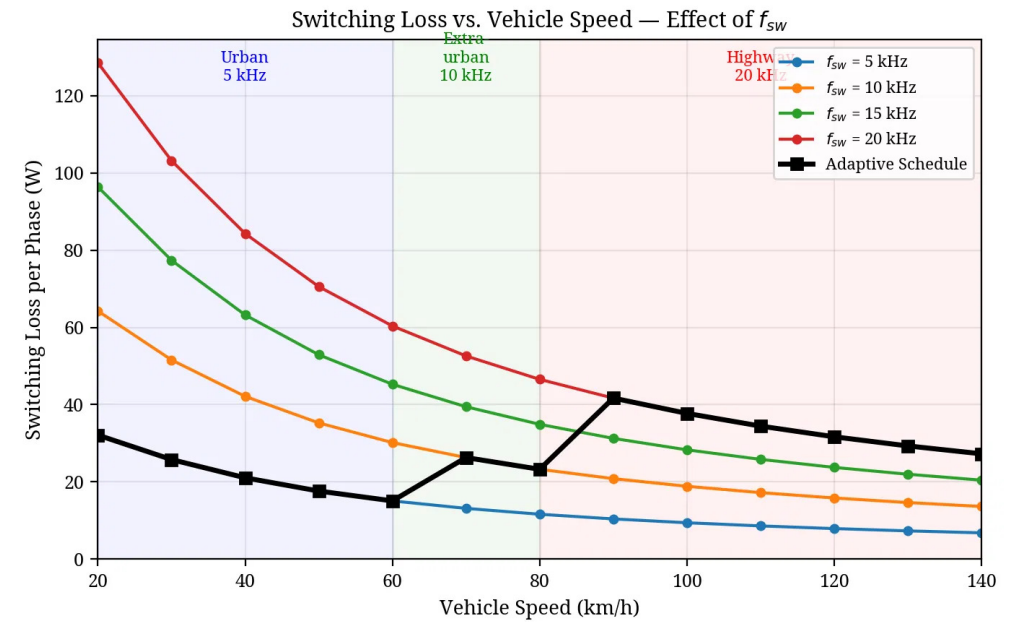
- **Capacitance (680 μ F):** Keeps ripple within 5% (20V on 400V) at 5 kHz min switching.
- **Film vs. Electrolytic:** Automotive-grade metallized polypropylene chosen.
- **Key Advantages:** Very low ESR (<5 m Ω), long life, handles high-frequency ripple.

Vehicle Architecture Mapping

- **Drivetrain Parameters:** Single-speed reduction gear ($G = 8.19$), tyre radius ($r = 0.315$ m), and 8-pole PMSM ($p = 4$).
- **Fundamental Frequency (f_e):** Directly proportional to vehicle speed.

$$f_e = (v / r) \times (G / 2\pi) \times p$$

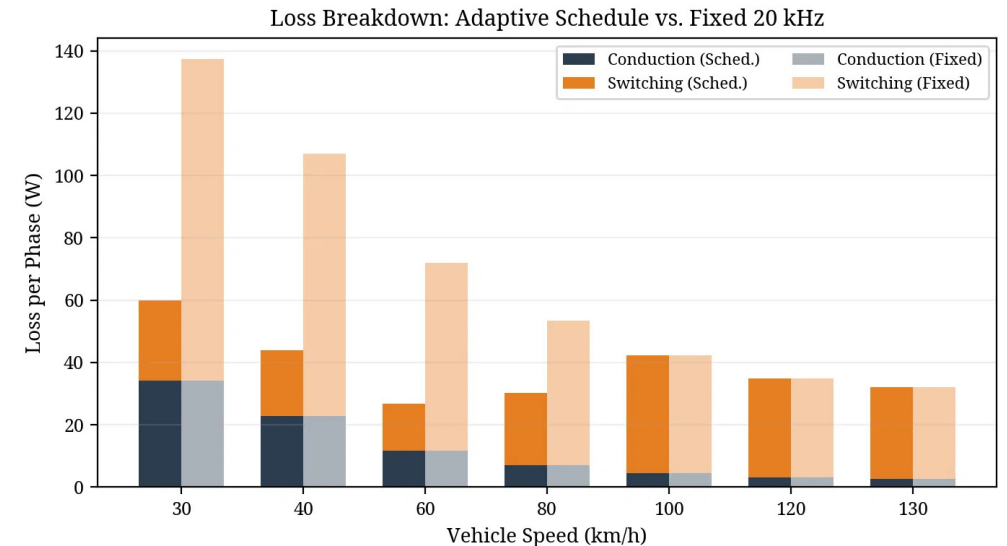
- **Urban Driving (40 km/h):** Low frequency (184 Hz) but high phase current (136 A) for acceleration. Represents the highest current stress.
- **Highway Cruising (120 km/h):** High frequency (552 Hz) but significantly lower sustained current (51 A) due to increased load impedance.



Switching Loss Analysis

$$P_{sw} = N_c \times (f_{sw} / \pi) \times E_{sw_ref} \times (I_{pk} / I_{ref}) \times (V_{dc_sw} / V_{ref})$$

- **Graovac-Purschel Method:** Analytical method to capture drive-frequency effects.
- **Applicability to SiC:** Topology-dependent math; works for SiC by using device curves.
- **Datasheet Calibration:** Uses CAB450M12XM3 parameters ($E_{on} = 25.4 \text{ mJ}$, $E_{off} = 7.51 \text{ mJ}$ @ 450A/600V).
- **Fixed 20 kHz Issue:** At 20 kHz switching losses dominate at low speeds due to high phase currents (~136A).



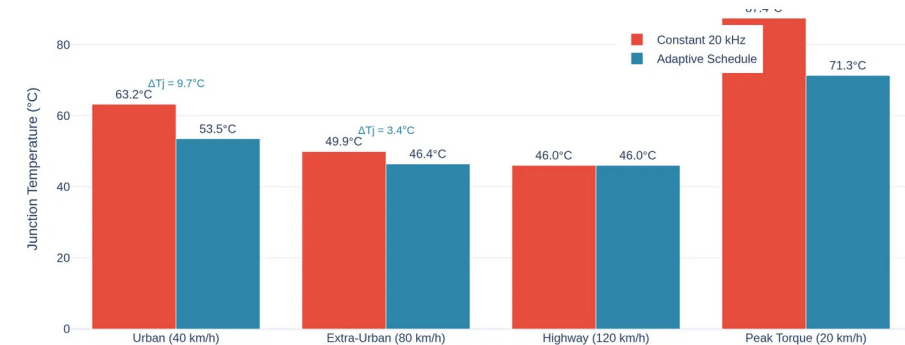
Simulated Adaptive Switching Frequency

An adaptive switching frequency schedule was developed and fully implemented within the time-domain simulation. The core methodology relies on maintaining a sufficient frequency modulation ratio ($m_f = f_{sw}/f_e$) to ensure waveform quality while minimizing losses.

Urban Driving (Low f_e)

Because f_e is low (184 Hz), the controller dynamically reduces f_{sw} to 5 kHz while maintaining a high modulation ratio ($m_f \approx 27$). This yields massive switching loss savings exactly when the phase current is highest.

Junction Temperature at Key Operating Points



Thermal Impact of Adaptive Schedule

- Thermal Modeling:** Junction temperature (T_j) was evaluated using a 4-layer Foster thermal network derived directly from the CAB450M12XM3 datasheet.
- Steady-State Reduction:** The adaptive schedule reduces T_j by 16.2°C during peak torque conditions (20 km/h) and by 9.7°C during urban driving compared to a fixed 20 kHz baseline.
- Transient Thermal Cycling:** Over a dynamic drive cycle, the adaptive schedule reduces the thermal cycling amplitude (ΔT_j) by 53% (from 16.4°C to 7.7°C).
- Reliability Gain:** According to the Coffin-Manson fatigue model, this massive reduction in thermal cycling amplitude directly and significantly extends the power module's operational lifetime.

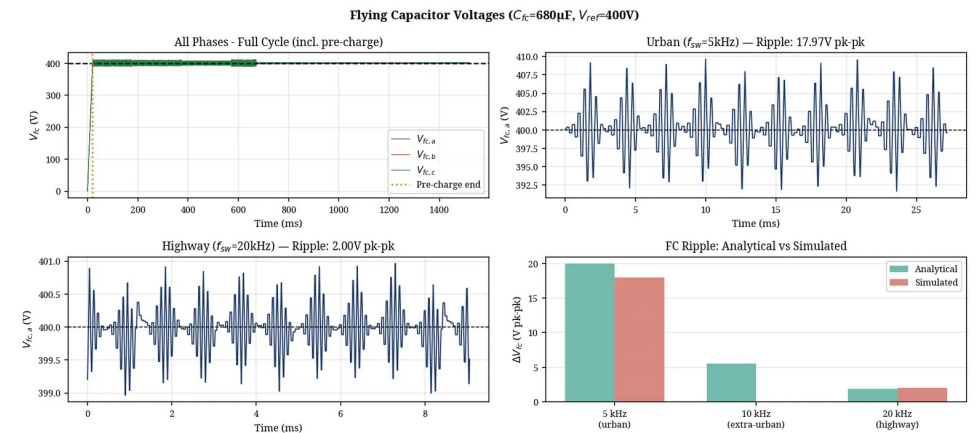


Fig. 4. Time-domain simulation validating FC voltage balancing, a prerequisite for stable thermal operation.

Simulation Results

FC Voltage Balancing

- **Pre-charge Sequence:** A 20ms linear ramp is implemented to prevent inrush currents during startup.
- **Natural Balancing:** PD-PWM successfully keeps the flying capacitor voltage tightly regulated at exactly 400V.
- **Ripple Budget:** The peak-to-peak voltage ripple is 18.0V, successfully meeting the strict 5% (20V) design constraint.

THD Validation

- **Harmonic Performance:** The 3-level output inherently reduces the voltage step size, significantly improving power quality.
- **Urban Driving:** At a 5 kHz switching frequency, the output Total Harmonic Distortion (THD) is 1.92%.
- **Highway Driving:** At a 20 kHz switching frequency, the output THD is 1.13%.

Thermal Analysis

- **Foster Network:** A 4-layer thermal model accurately captures the transient junction temperature dynamics.
- **Peak Temperature:** The junction temperature (T_j) peaks at only 50.1°C under the adaptive schedule.
- **Safety Margin:** This provides a massive 124.9°C safety margin to the 175°C semiconductor limit.

Conclusion

Key Findings

- **Topology Validation:** The 800V 3-Level Flying Capacitor topology successfully mitigates dv/dt stress and improves harmonic performance compared to 2L-VSI.
- **Loss Reduction:** The simulated adaptive switching frequency schedule reduces urban switching losses by **75%** (from 84.3W to 21.1W per phase).
- **Thermal Improvement:** The adaptive schedule reduces thermal cycling amplitude by **53%**, significantly extending the lifetime of the SiC power modules.

Limitations & Future Work

- **Ideal Switching:** The current time-domain simulation employs ideal switching transitions.
- **Unmodeled Parasitics:** The model does not currently account for device voltage drops, discrete dead-time distortion, DC bus ripple, or stray inductance.
- **Future Enhancements:** Incorporating these non-idealities in future iterations would yield more conservative and hardware-accurate THD and loss estimates.